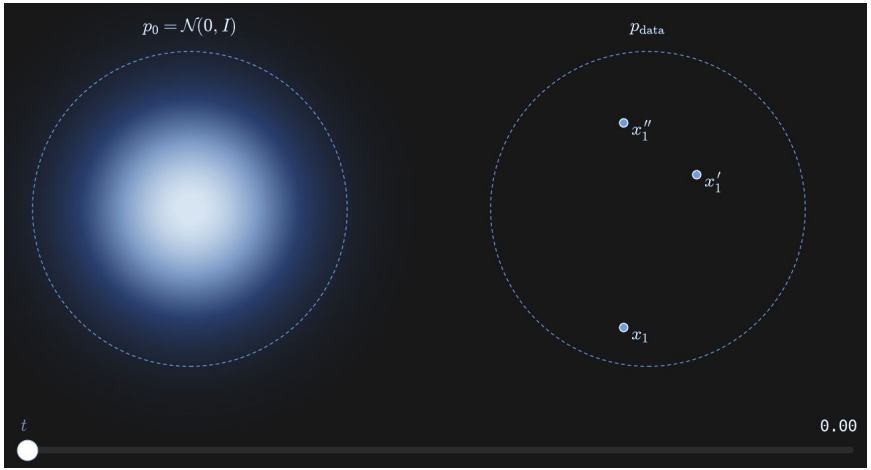
Flow matching

and something around

Мотивация

Генерация - это перенос распределения.

$$p_0 = \mathcal{N}(0, I) \quad p_1 \approx p_{\text{data}}$$



p_{data} true (but unknown) density of the data

p_1 density, that concentrates probability mass around the known samples

Глобально наша задача - смоделировать $\frac{\partial p_t}{\partial t}$

https://diffusion.fyi/tutorials/flow-matching

Перенос распределения

Th1 (Continuity equation) - an equation that describes the transport of some quantity.

$$\frac{\partial p_t}{\partial t} = -\nabla (p_t v_t)$$

$dx_t = v_t \cdot dt$ переходим к моделированию поля скоростей

Утв1 (Optimal transport between Gaussian distributions)

Для двух гауссовских распределений ОТ относительно квадратичной функции стоимости $c(x, y) = \|x - y\|^2$ является аффинным, то есть имеет вид $T(x) = Ax + b$

Утв1 (Optimal interpolation between Gaussian distributions)

Оптимальный непрерывный перенос между двумя гауссианами задается линейной интерполяцией x_0 в $T(x_0)$

$$x_t = (1-t)x_0 + tT(x_0) = (1-t)x_0 + t(\sigma x_0 + x_1) = [\sigma = 0] = (1-t)x_0 + tx_1$$

$$\frac{\partial x_t}{dt} = x_1 - x_0 = v^*$$

Лосс. Обучение. Инференс

Выбор лосса

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t\sim U[0,1],\ x_0\sim p_0,\ x_1\sim p_1}\left[\left\|v_\theta(x_t,t) - (x_1 - x_0)\right\|^2\right]$$

Модель, учась в conditional режиме, неявно выучивает и unconditional velocity field

$$v_\theta(x_t,t) = \mathbb{E}_{x_1\sim p_{\text{data}(x_1|x_t)}}\left[v_\theta(x_t|x_1)\right]$$

Algorithm 1 Training

Input: Data distribution p_{data}, velocity model v_θ

repeat
  Sample $x_0 \sim \mathcal{N}(0, I)$
  Sample $x_1 \sim p_{\text{data}}$
  Sample $t \sim \mathcal{U}(0, 1)$
  $x_t \leftarrow (1-t)x_0 + tx_1$
  $\mathcal{L} \leftarrow \|v_\theta(x_t, t) - (x_1 - x_0)\|_2^2$
  $\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}$
until convergence

Algorithm 2 Inference (Euler sampling)

Input: Trained velocity model v_θ, number of steps N

Sample $x_0 \sim \mathcal{N}(0, I)$
$\Delta t \leftarrow 1/N$
for $k = 0, \ldots, N-1$ **do**
  $t_k \leftarrow k\Delta t$
  $x_{k+1} \leftarrow x_k + \Delta t\, v_\theta(x_k, t_k)$
end for
return x_N

https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html

Reflow

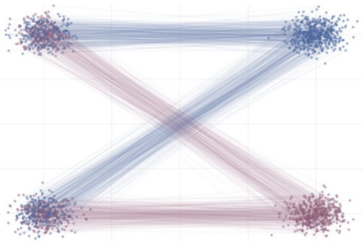


Figure 18: Realizations of conditional paths following conditional vector field $u_t(x \mid x_1)$ from (MoG-to-MoG). Paths are highlighted by the sign of the 2nd vector component.

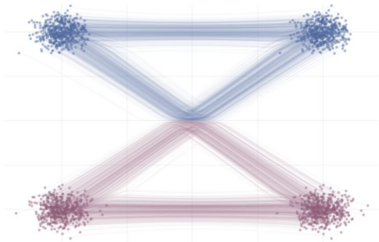
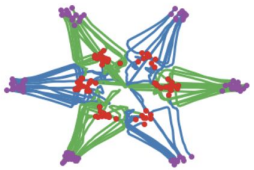


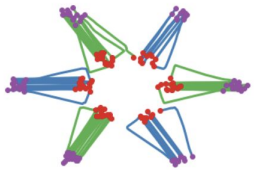
Figure 19: Realizations of marginal paths following the marginal vector field $u_t(x)$ from (MoG-to-MoG). Paths are highlighted by the sign of the 2nd vector component.

Чем более прямая траектория - тем лучше.



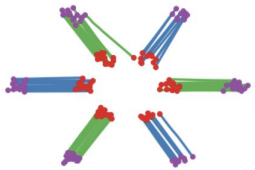
(a)The 1st rectified flow $\boldsymbol{Z}^1$

$\boldsymbol{Z}^1 = \texttt{RectFlow}((X_0, X_1))$



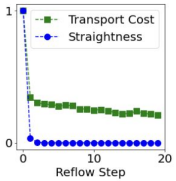
(b) Reflow $\boldsymbol{Z}^2$

$\boldsymbol{Z}^2 = \texttt{RectFlow}((Z_0^1, Z_1^1))$



(c) Reflow $\boldsymbol{Z}^3$

$\boldsymbol{Z}^3 = \texttt{RectFlow}((Z_0^2, Z_1^2))$



(d) Transport cost, Straightness

Algorithm 3 Reflow: Training a 2-Rectified Flow

Input: Pretrained velocity field $v_{\theta^{(1)}}$

Input: Number of synthetic pairs M, ODE steps N

Input: Learning rate η, training iterations K

Stage 1: Construct a new coupling

Freeze parameters $\theta^{(1)}$

$\mathcal{D}_{\text{reflow}} \leftarrow \emptyset$

for $i = 1, \ldots, M$ **do**

  Sample $x_0^{(i)} \sim \mathcal{N}(0, I)$

  $\hat{x}_1^{(i)} \leftarrow \text{ODESolve}(v_{\theta^{(1)}}, x_0^{(i)}, 0, 1, N)$

  $\mathcal{D}_{\text{reflow}} \leftarrow \mathcal{D}_{\text{reflow}} \cup \{(x_0^{(i)}, \hat{x}_1^{(i)})\}$

end for

Stage 2: Train a new velocity field

Initialize $\theta^{(2)} \leftarrow \theta^{(1)}$

for $k = 1, \ldots, K$ **do**

  Sample $(x_0, \hat{x}_1) \sim \mathcal{D}_{\text{reflow}}$

  Sample $t \sim \mathcal{U}(0, 1)$

  $x_t \leftarrow (1 - t)x_0 + t\hat{x}_1$

  $\mathcal{L} \leftarrow \|v_{\theta^{(2)}}(x_t, t) - (\hat{x}_1 - x_0)\|_2^2$

  $\theta^{(2)} \leftarrow \theta^{(2)} - \eta \nabla_{\theta^{(2)}} \mathcal{L}$

end for

return $v_{\theta^{(2)}}$

https://arxiv.org/pdf/2209.03003

DDPM

Algorithm 4 DDPM training

1: **repeat**
2: $x_0 \sim q(x_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\epsilon \sim \mathcal{N}(0, I)$
5: Take gradient descent step on
$\nabla_\theta \left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \, t \right) \right\|^2$
6: **until** converged

Algorithm 5 DDPM sampling

1: $x_T \sim \mathcal{N}(0, I)$
2: **for** $t = T, \dots, 1$ **do**
3: $z \sim \mathcal{N}(0, I)$ if $t > 1$, else $z = 0$
4: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$
5: **end for**
6: **return** x_0

Исходная формула прямого процесса $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon \longrightarrow x_t = a_t x_0 + \sigma_t \epsilon$

Параметризации

small noise (t ~0)

$a_t \approx 1 \; \sigma_t \approx 0 \; x_t \approx x_0$

$$x_0 = \frac{x_t - \sigma_t \epsilon}{a_t}$$

$$x_t = \cos \varphi_t \, x_0 + \sin \varphi_t \, \epsilon$$

$$\frac{dx_t}{d\varphi_t} = v_t \longrightarrow v_t = a_t \epsilon - \sigma_t x_0$$

адаптивно меняет свой target
в зависимости от SNR

big noise (t ~T)

$a_t \approx 0 \; \sigma_t \approx 1 \; x_t \approx \epsilon$

$$\epsilon = \frac{x_t - a_t x_0}{\sigma_t}$$

$$\text{SNR}(t) = \frac{a_t^2}{\sigma_t^2} = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}$$